

Technical Datasheet: Cloud Data Platform (AWS + Databricks)

Version: 1.0 · **Date:** 29 Sep 2026 · **Type:** Data Engineering portfolio project, 100% cloud, real data

1. Objective

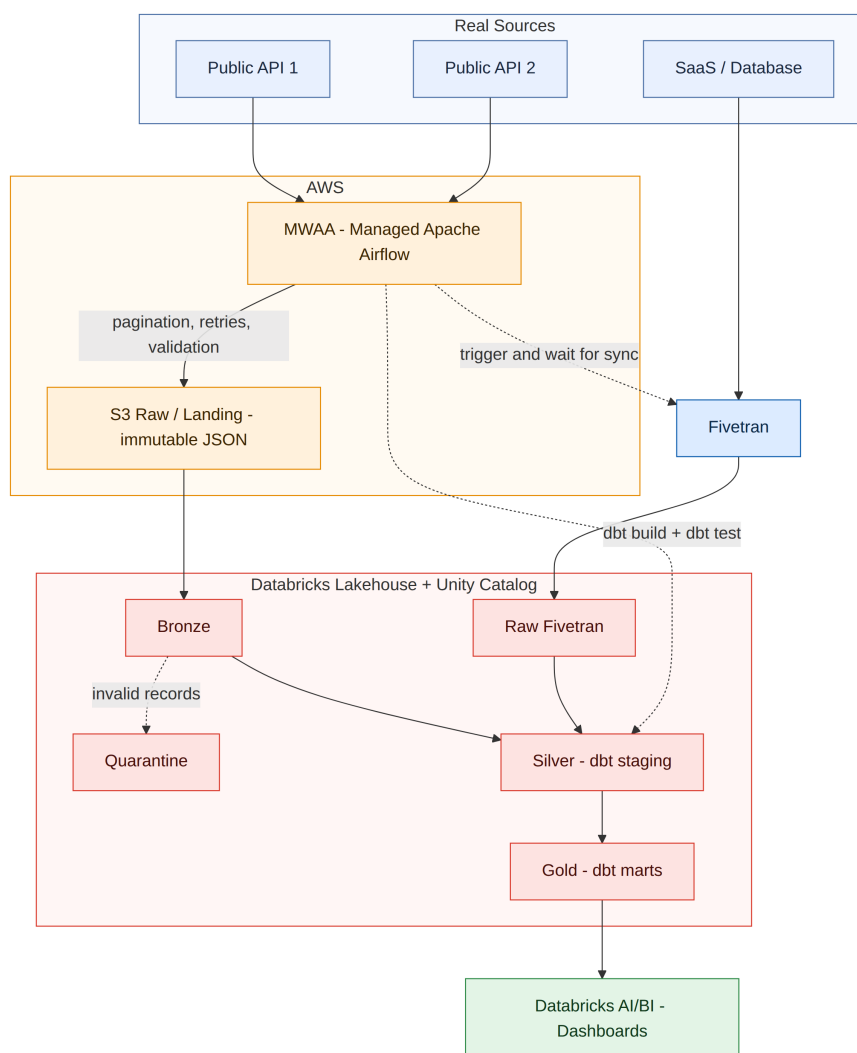
Build a modern end-to-end data platform with **real sources** (public APIs and one SaaS/DB connector), controlled ingestion, a medallion architecture on Databricks, dbt transformations, Unity Catalog governance, infrastructure as code and CI/CD.

"The sources are real; the pipeline that ingests, transforms and governs them is real too."

2. Architecture Principles

- **Real data from day one.** No simulators as a source; `datasets/` is only used for test fixtures.
- **Two ingestion strategies:** custom code (Python + Airflow) for APIs, and Fivetran for managed connectors.
- **Immutable raw layer on S3:** no overwrites; every ingestion writes a new object per `ingest_date`.
- **Core independent of Fivetran:** AWS + Databricks + dbt + Airflow work on their own; Fivetran is an additional path.
- **Minimal infrastructure:** only the AWS services that are actually needed.
- **Everything as code:** Terraform, dbt, DAGs and workflows are versioned on GitHub.

3. Architecture Diagram



Solid arrows represent data flow; dashed arrows represent orchestration or diversion of invalid records.

4. Technology Stack

Layer	Technology	Role
Cloud	AWS	Base infrastructure
Object storage	Amazon S3	Immutable landing/raw zone, MWAA DAGs
Orchestration	Amazon MWAA (managed Airflow)	Schedules, dependencies, retries, alerts
API ingestion	Python + Airflow	Pagination, rate limiting, retries, validation, raw JSON
SaaS/DB ingestion	Fivetran	Managed connectors, direct write to Databricks
Lakehouse	Databricks (SQL Warehouse)	Processing and SQL
Storage format	Delta Lake	Transactional tables
Governance	Unity Catalog	Catalog, permissions, lineage, auditing
Transformation	dbt-core + dbt-databricks	Staging, marts, tests, docs
Data quality	dbt tests + dbt-utils	Data tests
History	dbt snapshots	SCD2
BI	Databricks AI/BI	Business and data quality dashboards
IaC	Terraform	AWS + Databricks
CI/CD	GitHub Actions + OIDC	Tests, plan/apply, deployment
Secrets	AWS Secrets Manager	API keys and credentials
Monitoring	Airflow + CloudWatch + Databricks	Execution and health

5. Component Description

5.1 Sources

- **Public APIs** (exchange rates, weather, economy, transport, etc.): two APIs with different structures, to show this is not a trivial single integration.
- **SaaS/DB via Fivetran**: one real source with a ready-made connector (for example PostgreSQL, HubSpot or Salesforce).

5.2 Ingestion

Python + Airflow (APIs): each task calls the API, handles pagination and rate limits, applies retries, validates the schema and writes the JSON to S3 using [S3Hook](#).

Fivetran: writes directly to Delta tables managed by Unity Catalog, without an artificial hop through the landing bucket. Airflow **coordinates** it (triggers and waits for the sync through the Fivetran API); it does not replace it.

5.3 Amazon S3: Data Lake

Immutable landing zone, partitioned by ingestion date:

```
s3://project-landing/
├── api/
│   ├── exchange_rates/ingest_date=2026-09-29/response.json
│   └── weather/ingest_date=2026-09-29/response.json
```

Versioning enabled and SSE-S3 encryption by default.

5.4 MWAA: Orchestration

Airflow managed by AWS. DAGs and `requirements.txt` live in S3; logs go to CloudWatch. Main DAG `daily_pipeline`:

`API ingestion > check S3 > Fivetran sync > wait/validate > dbt build > dbt test > data quality > marts available`

5.5 Databricks Lakehouse: Medallion Architecture

Layer	Content
Bronze	Data as close as possible to the source, with technical metadata (<code>_batch_id</code> , <code>_ingested_at</code> , <code>_source_file</code> , <code>_source_system</code> , <code>_batch_date</code>)
Silver	Parsing, types, deduplication, normalization, business rules (dbt staging)
Gold	Analytical models: dimensions and facts (dbt marts)
Quarantine	Records rejected by validations
Snapshots	SCD2 history

5.6 dbt

Does not handle the main ingestion; it handles the Bronze > Silver > Gold transformation, plus tests, documentation, snapshots and artifacts.

5.7 Unity Catalog

Structure `catalog.schema.table` (e.g. `dwh_prod.bronze`, `.silver`, `.gold`, `.quarantine`, `.snapshots`).

- **Permissions:** pipeline with write access, BI with read access, developers as needed.
- **Lineage:** source > bronze > silver > gold > dashboard (removes the need for Marquez).
- **S3 access:** through storage credentials and external locations.

5.8 Databricks AI/BI

- **Business dashboard:** revenue by month/country/product, units, customers, average order value (or equivalents for the chosen APIs).
- **Data quality dashboard:** pipeline status, rows ingested/rejected, failed tests, latest batch, freshness.

5.9 Terraform

```
infra/terraform/  
├─ aws/      (s3, iam, mwaa, networking, budgets)  
└─ databricks/ (workspace, catalogs, schemas, grants, warehouses)
```

5.10 GitHub Actions

Event	Actions
Pull Request	tests, <code>dbt parse</code> , lint, gitleaks, <code>terraform plan</code>
Merge to <code>main</code>	<code>terraform apply</code> , Databricks deployment, <code>dbt build</code>

AWS authentication through **OIDC**, with no access keys stored in GitHub.

6. AWS Services

Essential	Optional (only if needed)	Out of scope
S3, IAM, MWAA, VPC, CloudWatch Logs, Secrets Manager, AWS Budgets	KMS, SNS, EventBridge, CloudTrail	Redshift, Glue, EMR, Athena, Kinesis, Lambda, DynamoDB, ECR, ECS, RDS

7. Decisions and Risks

Topic	Decision / note
Is Fivetran mandatory?	No. The Databricks destination requires Premium or higher; if the cost does not make sense, the project stays fully functional with Airflow alone.
Do all sources go through S3?	No. APIs go through S3 (immutable raw); Fivetran writes directly to Databricks.
KMS?	Not at the start; SSE-S3 is enough for a portfolio project.
Costs	AWS Budgets from day one; MWAA and the SQL Warehouse are the most expensive items, so shut resources down when not in use.

8. Repository Structure

```
cloud-data-platform/  
├─ ingestion/          (api/, fivetran/)  
├─ orchestration/     (dags/)  
├─ dbt_project/        (models/, snapshots/, tests/, macros/)  
├─ infra/terraform/    (aws/, databricks/)  
├─ docs/  
├─ tests/  
├─ .github/workflows/  
├─ README.md  
└─ AGENTS.md
```

9. Suggested Build Order

1. S3 + IAM + Unity Catalog (storage credential, external location)
2. First API > S3 > Bronze
3. dbt (staging + marts + tests)
4. MWAA and the main DAG
5. Fivetran (second ingestion path)
6. Terraform and GitHub Actions
7. AI/BI dashboards and documentation